

Synth Workflow: From Concept to Prototype

This guide is for developing a workflow for designing your device.

Step 1: Define the Instrument Concept

- Choose a sensor input.
- Decide which musical parameter the sensor will control (pitch, volume, filter, effects).
- Select a sound source (sine, square, saw, noise).

Step 2: Develop the Audio Signal Schematic

- Create a diagram showing the flow of information from sensor to sound output.
- Include sensor input, mapping/processing, sound generation, and speaker/headphone output.
- Use the framework provided on github to show signal flow.

Step 3: Build a PlugData Patch

- Recreate the schematic using PlugData objects.
- Use sliders, knobs, or buttons to simulate sensor behaviour.
- Create the oscillator and audio output chain.
- Test the sound and interaction design.

Step 4: Translate the Patch into Python Components

- Identify a CircuitPython equivalent for each PlugData object.
- Input object becomes sensor reading code.
- Mapping stage becomes frequency or parameter calculations.
- Oscillator becomes Synthio notes and waveforms.

Step 5: Develop Python Script Prototypes

- Prototype 1: Generate a basic sound.
- Prototype 2: Read sensor values and print them.
- Prototype 3: Map sensor values to sound parameters.
- Prototype 4: Combine sensor input and audio output into a working instrument.

Step 6: Compare All Three Assessment Components

- Audio Signal Schematic explains how the instrument works.
- PlugData Patch demonstrates the interaction and sound qualities.
- Python Prototype implements the design on the Pico using CircuitPython.

Installing CircuitPython on a Pico

This section is for students who have purchased their own Raspberry Pi Pico and need to prepare it for the synth project.

Before You Start

- Raspberry Pi Pico (RP2040)
- USB data cable
- Computer (Windows, macOS)
- Thonny installed
- Internet connection

Step 1: Download CircuitPython

- Visit the CircuitPython download page for Raspberry Pi Pico.
- https://circuitpython.org/board/raspberry_pi_pico/
- Download the latest UF2 file for your board.

Step 2: Enter Bootloader Mode

- Disconnect the Pico.
- Hold the BOOTSEL button.
- Connect the Pico while holding BOOTSEL.
- Release BOOTSEL when the RPI-RP2 drive appears.

Step 3: Install CircuitPython

- Drag the downloaded UF2 file onto the RPI-RP2 drive.
- Wait for the board to reboot automatically.
- Verify that a drive named CIRCUITPY appears.

Step 4: Install a Code Editor

- Open Thonny Editor.

Step 5: Test the Installation

- Open the CIRCUITPY drive.
- Check that boot_out.txt and code.py are present.

Step 6: Copy github code onto the Pico

- Download and copy the entire contents of the Pico scripts onto your device
- [link](#)

Step 7: Install Libraries for the Synth Project

- Test the pico using any of the week 5 examples
- [link](#)

EX 1.1 Sensor Mapping and VCO Control

`code-temp-synth.py`

Sensor values are mapped to oscillator frequency, and this affects the pitch range of a synthesizer.

In the Pico synth, the temperature value acts like a digital control voltage. The sensor reading becomes the control signal that changes the oscillator frequency.

- Sensors provide control data.
- Mapping converts sensor data into useful musical values.
- Larger frequency ranges create dramatic pitch changes.
- The same mapping process can be used with any other sensor.

Sensor Mapping.

Sensors usually produce values that are not musically useful by themselves. Mapping converts one range of values into another range. In a synthesizer, mapping is commonly used to convert sensor readings into frequency values for a VCO.

How Mapping Affects Pitch Range

Example 1: Small Pitch Range

20°C → 220 Hz

35°C → 440 Hz

The pitch changes over one octave, producing a relatively gentle musical response.

Example 2: Large Pitch Range

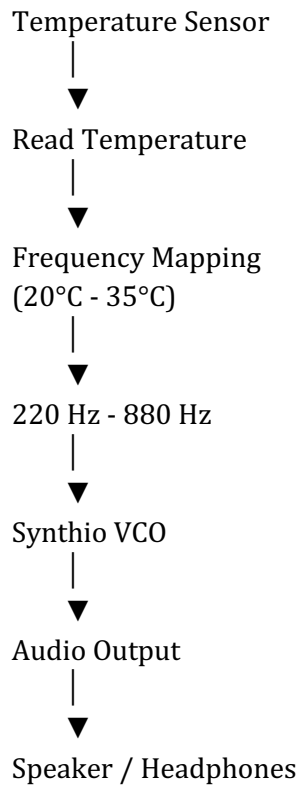
20°C → 220 Hz

35°C → 1760 Hz

The pitch changes across several octaves, producing a much more dramatic response.

Changing the minimum and maximum frequency values changes how sensitive the instrument feels and how much of the musical range is available to the performer.

Sensor mapped to a VCO



Traditional Analogue Synth Comparison

